

A011 · Elucidating the role of TaVER2 and Rice orthologs in Xylan biosynthesis

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The plant cell wall is a dynamic structure providing strength, shape and protection to the plant cell. Heteroxylan is an important cell-wall polysaccharide in grasses such as rice and wheat where it functions in structural integrity, growth, biomass quality and stress responses. TaVER2, a vernalization-related protein with jacalin and dirigent domains, was a candidate regulator in wheat that may influence xylan production by association with the xylan synthase complex. However, its biological role in living plants, especially in rice, has not been completely demonstrated. In this study, the role of TaVER2 and its rice orthologs Os12g0198700 and Os12g0247700 in heteroxylan biosynthesis and plant growth and development was investigated. Phylogenetic analysis, sequence comparison and 3D structural modelling indicated conservation of the jacalin and dirigent domains between TaVER2 and rice homologs, suggesting possible conservation of function. To test this hypothesis, we generated CRISPR/Cas9 knockout mutants for Os12g0198700 and transgenic rice lines overexpressing TaVER2. Initial observations show visible developmental differences compared to wild-type plants including reduced plant height and altered flowering time. These phenotypes indicate that the regulation of heteroxylan may affect plant growth and architecture in general. We also got the overexpression lines for the orthologs genes and characterizing them now. Molecular, biochemical and microscopic characterization of the mutant and overexpression lines is in progress. The role of these genes in cell-wall assembly is being investigated by analyzing heteroxylan content, sugar composition and cell-wall structure. Selected lines will also be grown under biotic stress conditions including Hessian fly infestation to determine if there are any correlations between heteroxylan regulation and plant defense.

Elucidating the Role of TaVER2 and Rice Orthologs in Heteroxylan Biosynthesis

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Abstract

The plant cell wall is a dynamic structure providing strength, shape and protection to the plant cell. Heteroxylan is an important cell-wall polysaccharide in grasses such as rice and wheat where it functions in structural integrity, growth, biomass quality and stress responses. TaVER2, a vernalization-related protein with jacalin and dirigent domains, was a candidate regulator in wheat that may influence xylan production by association with the xylan synthase complex. However, its biological role in living plants, especially in rice, has not been completely demonstrated. In this study, the role of *TaVER2* and its rice orthologs *Os12g0198700* and *Os12g0247700* in heteroxylan biosynthesis and plant growth and development was investigated. Phylogenetic analysis, sequence comparison and 3D structural modelling indicated conservation of the jacalin and dirigent domains between TaVER2 and rice homologs, suggesting possible conservation of function. To test this hypothesis, we generated CRISPR/Cas9 knockout mutants for *Os12g0198700* and transgenic rice lines overexpressing *TaVER2*. Initial observations show visible developmental differences compared to wild-type plants including reduced plant height and altered flowering time. These phenotypes indicate that the regulation of heteroxylan may affect plant growth and architecture in general. We also got the overexpression lines for the orthologs genes and characterizing them now. Molecular, biochemical and microscopic characterization of the mutant and overexpression lines is in progress. The role of these genes in cell-wall assembly is being investigated by analyzing heteroxylan content, sugar composition and cell-wall structure. Selected lines will also be grown under biotic stress conditions including Hessian fly infestation to determine if there are any correlations between heteroxylan regulation and plant defense.

Key Words: Rice, *TaVER2*, orthologs, heteroxylan, plant cell wall, CRISPR/Cas9, overexpression